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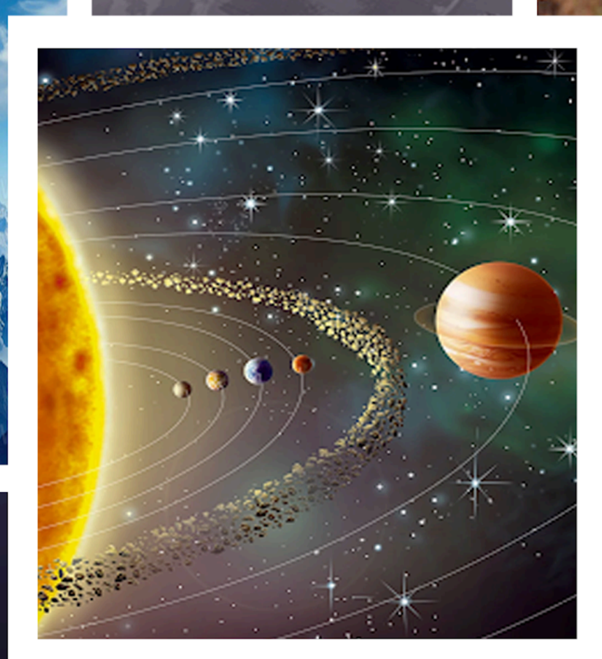
General Knowledge

World Geography

STUDY NOTES

Geomorphology: Landforms(River)



GEOMORPHOLOGY: LANDFORMS(RIVER)

Several related landforms together make up landscapes. Each landform has its own physical shape, size, and materials and is a result of the action of certain geomorphic processes and agents. Actions of most of the geomorphic processes and agents are slow, and hence the results take a long time to take shape. **Landforms once formed may change in their shape, size and nature slowly or fast due to the continued action of geomorphic processes and agents.** Due to changes in climatic conditions and vertical or horizontal movements of landmasses, either the intensity of processes or the processes themselves might change leading to new modifications in the landforms.

Fluvial Erosional Landforms

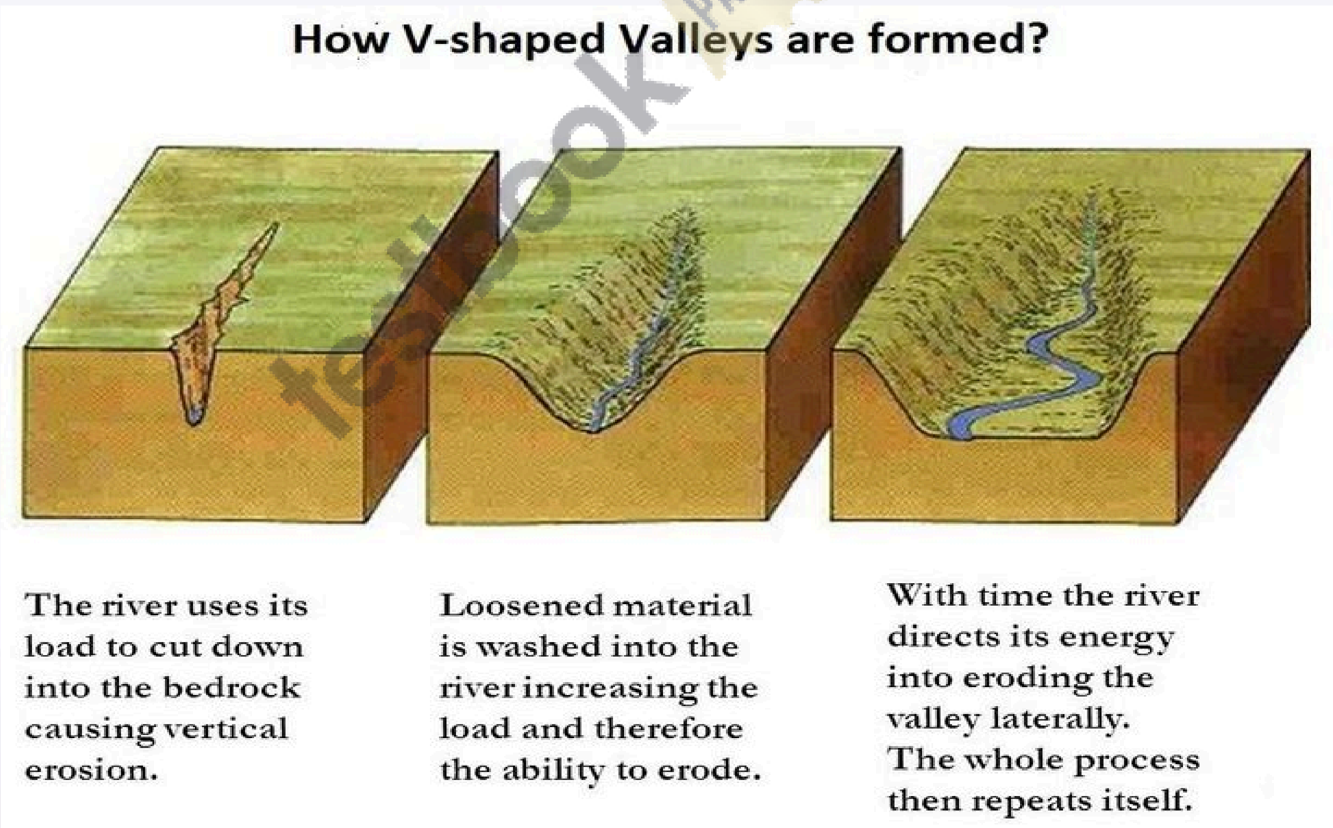
Fluvial Erosional Landforms are landforms created by the erosional activity of rivers. Various aspects of fluvial erosive action include:

- **Hydration:** the force of running water wearing down rocks.
- **Corrosion:** chemical action that leads to weathering.
- **Attrition:** river load particles strike, colliding against each other, and breaking down in the process.
- **Abrasion:** solid river load striking against rocks and wearing them down.
- **Downcutting (vertical erosion):** the erosion of the base of a stream (downcutting leads to valley deepening).
- **Lateral erosion:** the erosion of the walls of a stream (leads to valley widening).
- **Headward erosion:** erosion at the origin of a stream channel, which causes the origin to move back away from the direction of the stream flow, and so causes the stream channel to lengthen.
- **Braiding:** the main water channel splitting into multiple, narrower channels. A braided river, or braided channel, consists of a network of river channels separated by small, and often temporary, islands called braid bars. Braided streams occur in rivers with low slopes and/or large sediment loads.

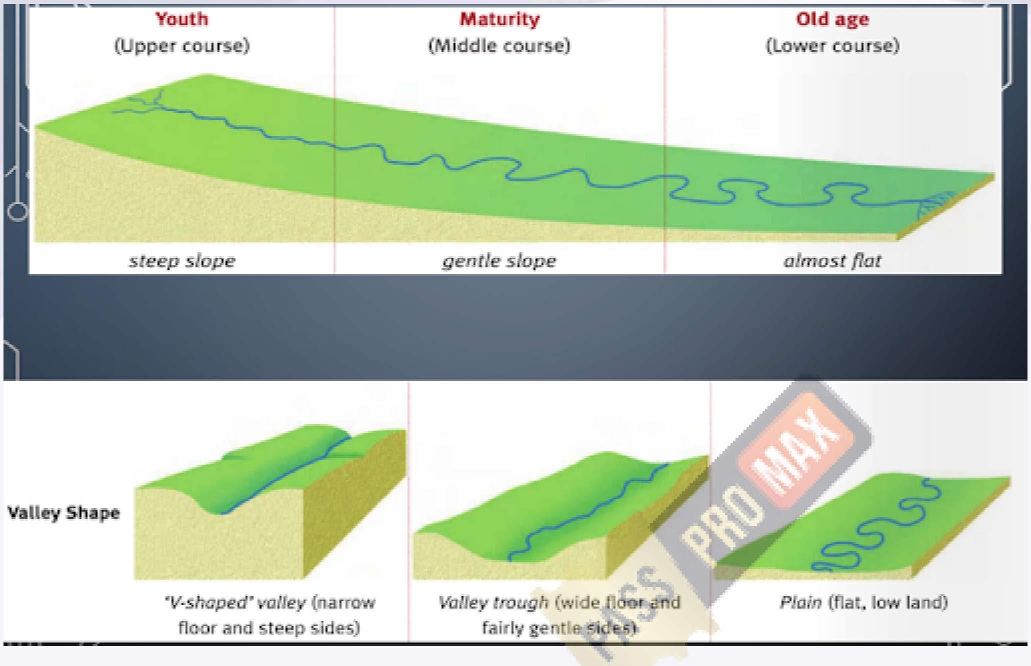
River Valley Formation

- **The extended depression on the ground through which a stream flows is called a river valley.** At different stages of the erosional cycle, the valley acquires different profiles.
- At a young stage, the valley is deep and narrow with steep wall-like sides and a convex slope.

- The erosional action here is characterized by a predominantly **vertical downcutting** nature.
- **The profile of the valley here is typically 'V' shaped.**
- **A deep and narrow 'V' shaped valley** is also referred to as a **gorge** and may result due to downcutting erosion or because of the recession of a waterfall (the position of the waterfall receding due to erosive action).
- **Most Himalayan rivers pass through deep gorges** (at times more than 500 metres deep) before they descend to the plains.
- An extended form of the gorge is called a **canyon.** The **Grand Canyon** of the **Colorado River in Arizona (USA)** runs for 483 km and has a depth of 2.88 km.
- A tributary valley lies above the main valley and is separated from it by a steep slope down which the stream may flow as a waterfall or a series of rapids.
- As the cycle attains maturity, the **lateral erosion** (erosion of the walls of a stream) becomes prominent and the valley floor flattens out (attains a 'V' to 'U' shape).
- **The valley profile now becomes typically 'U' shaped** with a broad base and a concave slope.



River Course



1) <u>**Youth**</u>

- Young rivers (A) close to their source tend to be fast-flowing, high-energy environments with rapid headward erosion, despite the hardness of the rock over which they may flow.
- Steep-sided **"V-shaped' valleys, waterfalls, and rapids** are characteristic features. E.g. Rivers flowing in the Himalayas.

2) <u>**Maturity**</u>

- Mature rivers (B) are lower-energy systems.
- Erosion takes place on the outside of bends, creating looping meanders in the soft alluvium of the river plain.
- Deposition occurs on the inside of bends and on the river bed.
- E.g. Rivers flowing in the Indo-Gangetic-Brahmaputra plain.

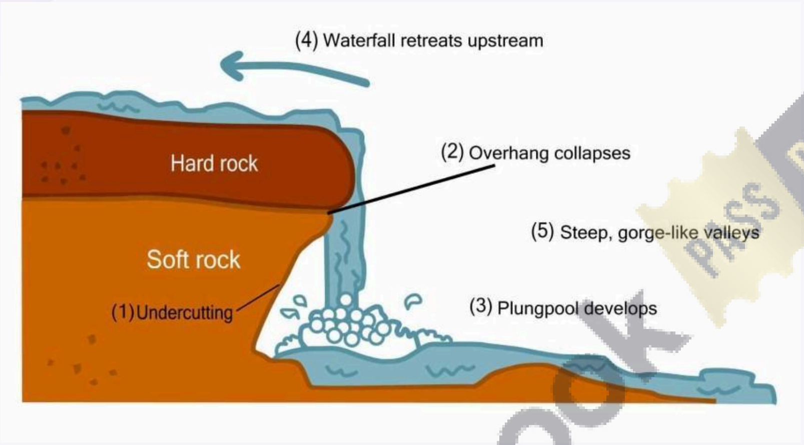
3) <u>**Old Age**</u>

- **At a river's mouth (C), sediment is deposited as the velocity of the river slows.**
- As the river becomes shallower more deposition occurs, forming temporary islands (Majuli, a river island in the Brahmaputra River, Assam is currently the world's largest river island) and braiding (e.g. braided channels of Brahmaputra river flood plain in Assam) the main channel into multiple, narrower channels.

- As the sediment is laid down, the actual mouth of the river moves away from the source into the sea or lake, forming a **delta**.
- E.g. Ganga-Brahmaputra delta.

Waterfalls

- A waterfall is simply the fall of an enormous volume of water from a great height.
- They are **mostly seen in** the **youth stage** of the river.
- Relative resistance of rocks, the relative difference in topographic reliefs, fall in the sea level and related rejuvenation, earth movements etc. are responsible for the formation of waterfalls.



- **Kunchikal Falls** (it is a cascade falls — falls with many steps) formed by Varahi river in Shimoga district, Karnataka is the highest waterfall in India (455 m).
- **Jog or Gersoppa Falls (253 m)** on Sharavathi river (a tributary of Cauvery), Karnataka is the second-highest plunge waterfall in India.
- **Angel Falls** in Venezuela is the world's highest waterfall, with a height of 979 metres and a plunge of 807 metres.

Potholes

- The small cylindrical depressions in the rocky beds of the river valleys are **called potholes.**
- **Potholing or pothole-drilling is the mechanism through which the fragments of rocks** when caught in the water eddies or swirling water start dancing circularly and grind and drill the rock beds.

- They thus form small holes which are gradually enlarged by the repetition of the said mechanism.

Terraces

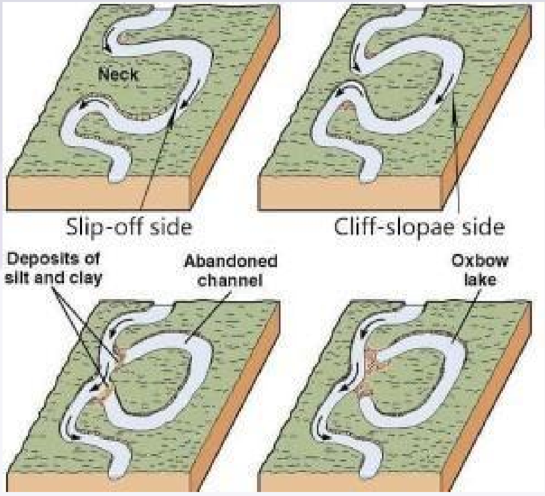
- Stepped benches along the river course in a floodplain are **called terraces.**
- Terraces represent the level of former valley floors and remnants of former (older) floodplains.

Gullies/Rills

- **Gulley is a water-worn channel, which is particularly common in semi-arid areas.**
- It is formed when water from overland flows down a slope, especially following heavy rainfall, and is concentrated into rills, which merge and enlarge into a gulley.
- The **ravines of Chambal Valley** in Central India and the **Chos of Hoshiarpur** in Punjab are examples of gullies.

Meanders

- A meander is defined as a pronounced curve or loop in the course of a river channel.
- The outer bend of the loop in a meander is characterised by intensive erosion and vertical cliffs and is called the **cliff-slope side**. This side has a concave slope.
- The inner side of the loop is characterized by deposition, a gentle convex slope, and is called the **slip-off side.**
- The meanders may be **wavy, horse-shoe type or oxbow type**.



Oxbow Lake

- Sometimes, because of intensive erosion action, the outer curve of a meander gets accentuated to such an extent that the inner ends of the loop come close enough to get disconnected from the main channel and exist as independent water bodies called oxbow lakes.
- These water bodies are converted into swamps over time.
- In the **Indo-Gangetic plains**, the southward shifting of the **Ganga** has left many oxbow lakes to the north of the present course of the Ganga.

- **Kanwar lake** is the **largest oxbow lake in India and in Asia**. It is a freshwater lake located in the Begusarai district and is the only Ramsar site in Bihar. It is currently facing the threat of running dry.

Peneplain

- This refers to an undulating featureless plain punctuated with low-lying residual hills of resistant rocks. It is considered to be an **end product of an erosional cycle**.

* Fluvial erosion, in the course of geologic time, reduces the land almost to base level (sea level), leaving so little gradient that essentially **no more erosion could occur**.

Fluvial Depositional Landforms

* **Fluvial Depositional Landforms are landforms created by the depositional activity of rivers.**
* **The depositional action of a stream is influenced by stream velocity and the volume of river load.** The decrease in stream velocity reduces the transporting power of the streams which are forced to leave some load to settle down.

Various landforms resulting from fluvial deposition are as follows:

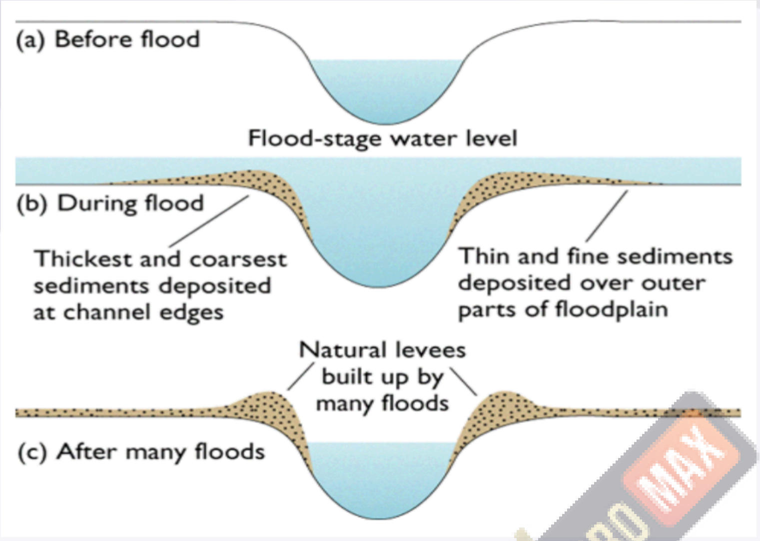
Alluvial Fans and Cones

* When a stream leaves the mountains and comes down to the plains, its velocity decreases due to a lower gradient.
* As a result, it sheds a lot of material, which it had been carrying from the mountains, at the foothills.
* This deposited material acquires a conical shape and appears as a series of continuous fans. These are called alluvial fans.
* Such fans appear throughout the **Himalayan foothills** in the north Indian plains.

Natural Levees

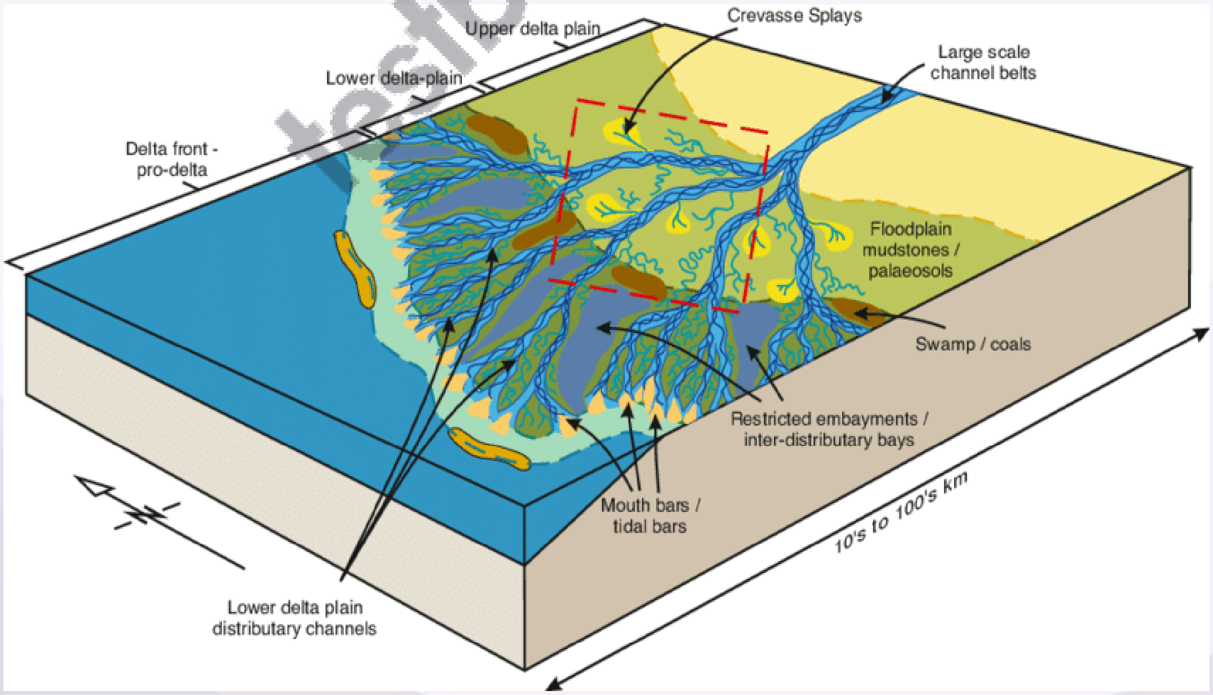
* These are narrow ridges of low height on both sides of a river, formed due to deposition action of the stream, appearing as natural embankments.

* These act as natural protection against floods but a breach in a levee causes sudden floods in adjoining areas, as happened in the case of the **Hwang Ho River of China**.



Delta

* A delta is a tract of alluvium at the mouth of a river where it deposits more material than can be carried away.
* The river gets divided into distributaries which may further divide and re-join to form a network of channels.



- A combination of two processes forms a delta:
  - The load-bearing capacity of a river is reduced as a result of the check to its speed as it enters a sea or lake, and
  - Clay particles carried in suspension in the river **coagulate** in the presence of salt water and are deposited.
- The finest particles are carried farthest to accumulate as bottom-set beds. Depending on the conditions under which they are formed, deltas can be of many types.

Type of Delta	Description	Example(s)
Arcuate Delta	Fan-shaped; bowed or curved with convex margin facing the body of water. Relatively coarse sediments. River activity is balanced with waves.	Nile River Delta (Egypt), Ganges Delta (India)
Bird's Foot Delta	Shaped like a bird's claw; formed when river flow is stronger than waves. Deposition of finer materials divides the river into smaller distributaries.	Mississippi River Delta (USA)
Cuspate Delta	Tooth-like shape; formed along straight shorelines with strong waves pushing sediments outward.	Tiber River Delta (Italy)
Estuarine Delta	Formed at the mouth of submerged rivers, depositing sediments along estuary sides.	Seine River Delta (France), Narmada and Tapi (India)
Lacustrine Delta	Formed when a river flows into a lake.	Lough Leanne River Delta (Ireland)
Truncated Delta	Modified or eroded by sea waves and ocean currents; eroded and dissected Deltas.	-
Abandoned Delta	Created when a river shifts its mouth, leaving the former Delta unused.	Yellow River Delta (China), Western Ganga Delta by Hooghly River (India)